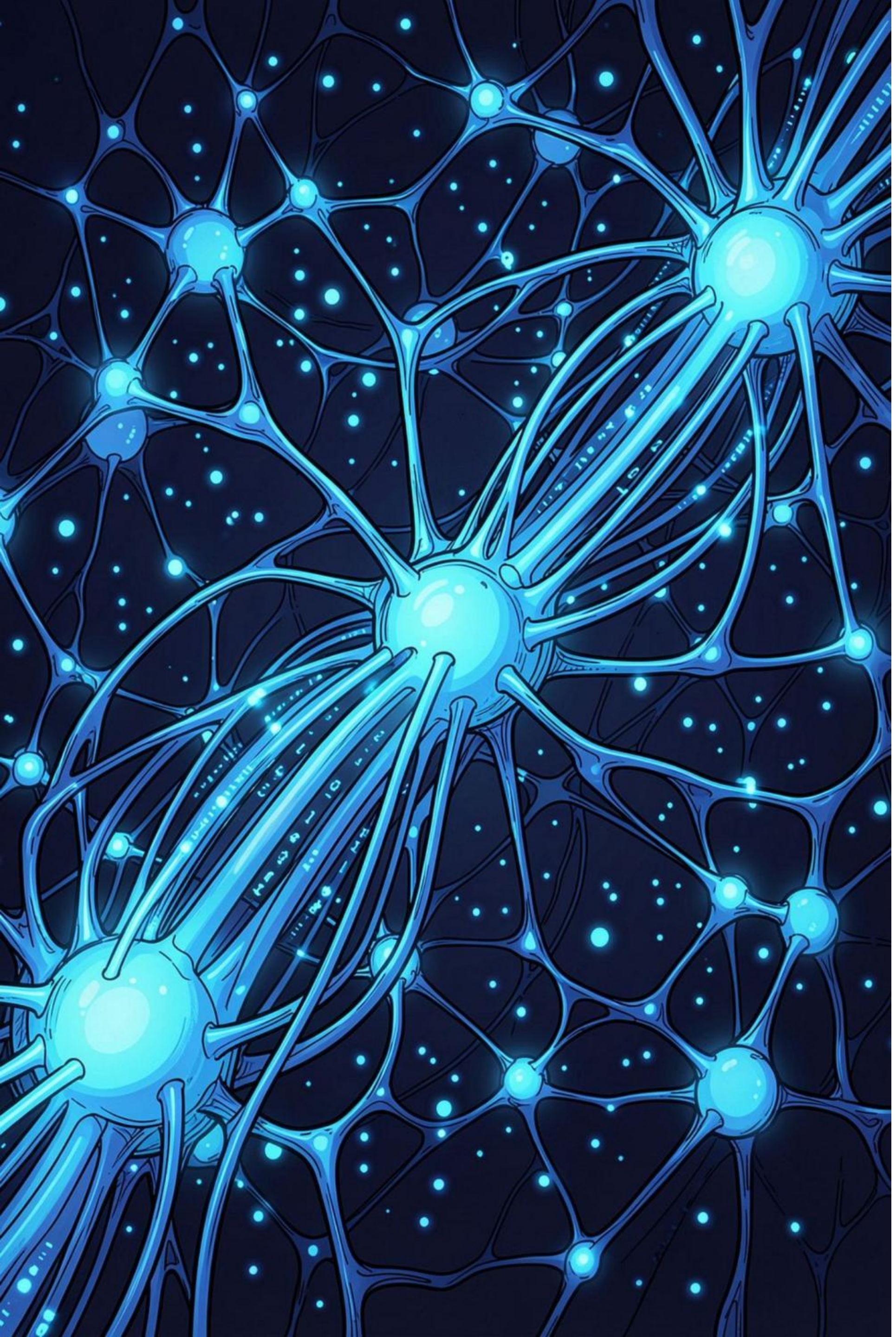


PII Detection and Anonymization

A Python pipeline for detecting and anonymizing Personally Identifiable Information (PII) in structured data and free text.

UNIVERSITY CS PROJECT

PYTHON · PII · NER

Ramin Hassani
Computer Science

The Problem: Detecting and Anonymizing PII

Datasets can contain PII in both structured fields and free text. The challenge is to automatically identify these entities and apply appropriate anonymization techniques without unnecessarily destroying useful information.

Structured PII

Predictable fields such as:

- Names
- Patient codes
- Tax codes
- Dates
- Age

Free-Text PII

PII can also appear inside natural language, where fixed rules are less reliable.

Example:

> “Anna lives in Rome.”

The system needs to identify **Anna** and **Rome** as entities.

Automation

Manual anonymization is difficult to scale across heterogeneous datasets.

The project therefore builds a pipeline that:

loads → detects → anonymizes → evaluates

The Dataset

Synthetic Diabetology Dataset

A fully synthetic Excel dataset simulating a clinical diabetology registry.

Key Structured PII Fields

Patient Code	First Name	Surname	Tax Code	Birth Date	Age at assessment
F87421	Anna	Rossi	RSS...	14/05/1952	74
F39778	Marco	Bianchi	BNC...	03/11/1961	64

1,337 rows

Patient records

103 columns

Clinical and administrative fields

System Architecture

A modular end-to-end pipeline processes any supported input format and produces an anonymized output dataset.

Input File → **Universal Loader** → **PII Detection** → **Anonymization** → **Anonymized Output**

Universal Loader

Accepts multiple input formats and converts them into a common internal representation:

- **Excel** (.xlsx / .xls)
- **CSV** (.csv)
- **JSON** (.json)
- **TXT** (.txt)

Structured Data

Rule-based / Regex Detection

- Patient codes
- Tax codes
- Dates & IDs

Free Text

- **NER Detection**
  Alternative detectors:
- spaCy
- Hugging Face
- GLiNER

The Universal Loader abstracts input-format differences. Detection supports two paths: rule-based regex for structured fields and NER for configured free-text fields. Anonymization then applies type-specific strategies before producing the final output.

PII Detection: Two Approaches

Structured Data — Regex Rules

Deterministic patterns for predictable, schema-bound fields.

Dates

Birth date, assessment date

Identifiers

Patient code, tax code, email

Phone Numbers

Standardized regex patterns

Free Text — Named Entity Recognition

Three alternative pretrained NER models for free-text data. Only one is selected per run for comparison.

Anna `PERSON` lives in Rome `LOCATION`.

Example: NR entity highlighting in a free-text sentence

spaCy

Fast, lightweight pretrained NER

Hugging Face XLM-RoBERTa

Davlan/xlm-roberta-base-ner-hrl

Multilingual transformer NER

GLiNER

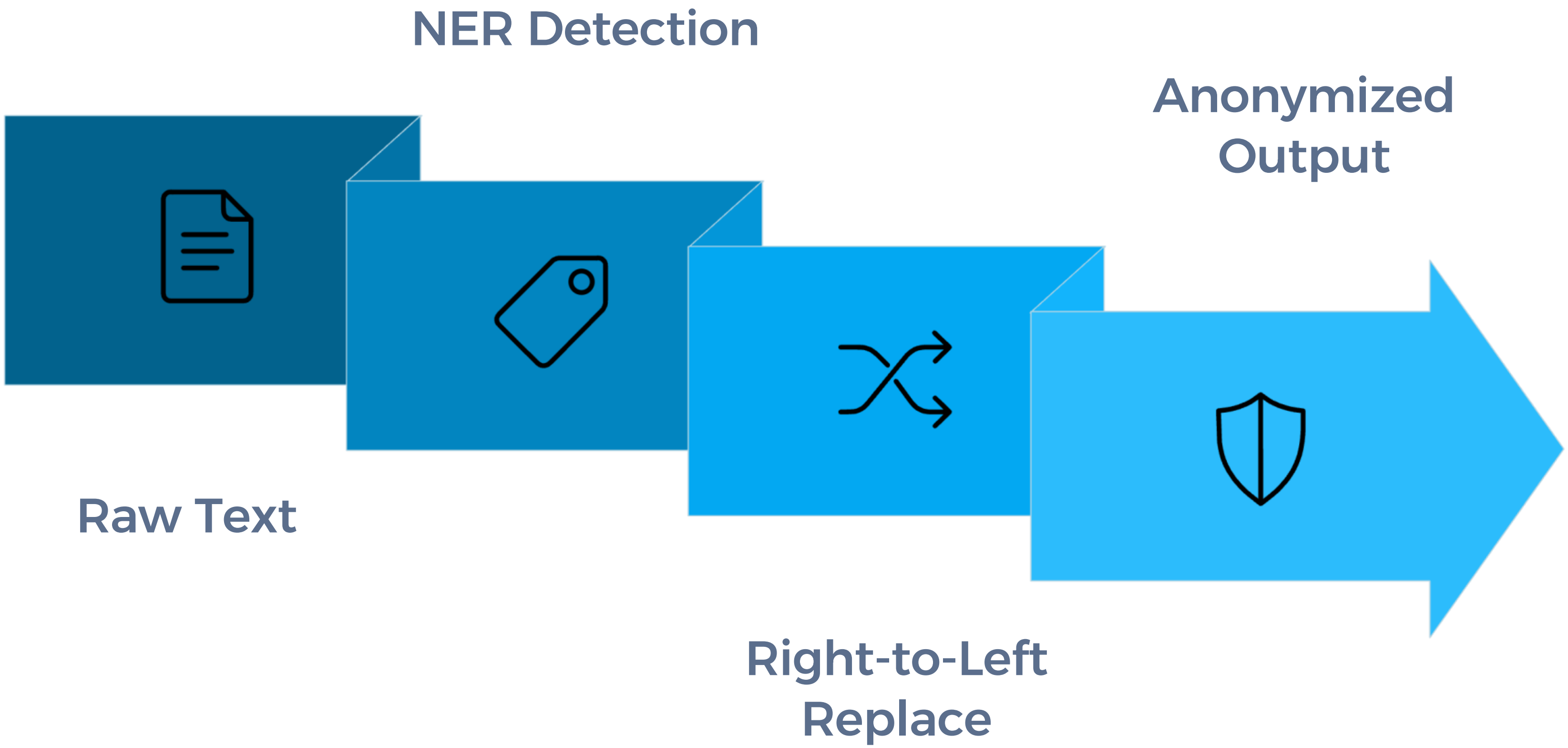
General-purpose NER model

Anonymization Strategies

Detection answers **"Where is the PII?"** Anonymization answers **"What should we do with it?"** Each PII type receives a tailored transformation.

PII Type	Strategy	Example Transformation
Patient Code	Pseudonymization	`ABC123 → PATIENT_0001`
First Name / Surname	Masking	`Mario Rossi → M**** R****`
Tax Code	Masking	`RSSMRA80A01H501U → RSS***********1U`
Dates	Generalization	`1980-05-12 → 1980`
Age	Range Generalization	`74 → 70–79`
Free-text entities	Placeholder Replacement	`"Mario" → [PERSON]`

Free-Text Anonymization Pipeline

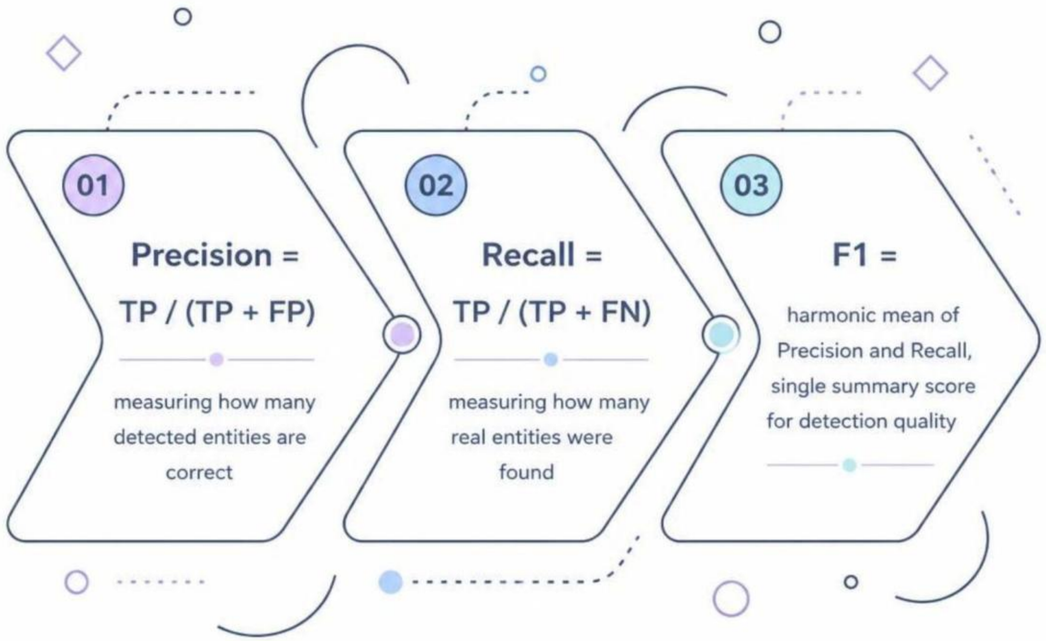


Before

After

Evaluation Methodology

Precision, Recall, F1



Benchmark: **36 examples**, **97 ground-truth entities**.

Strict Span-Based Matching

Ground truth provides the expected entity spans and types for each example.
A predicted entity is a **true positive** only when *both* conditions hold:

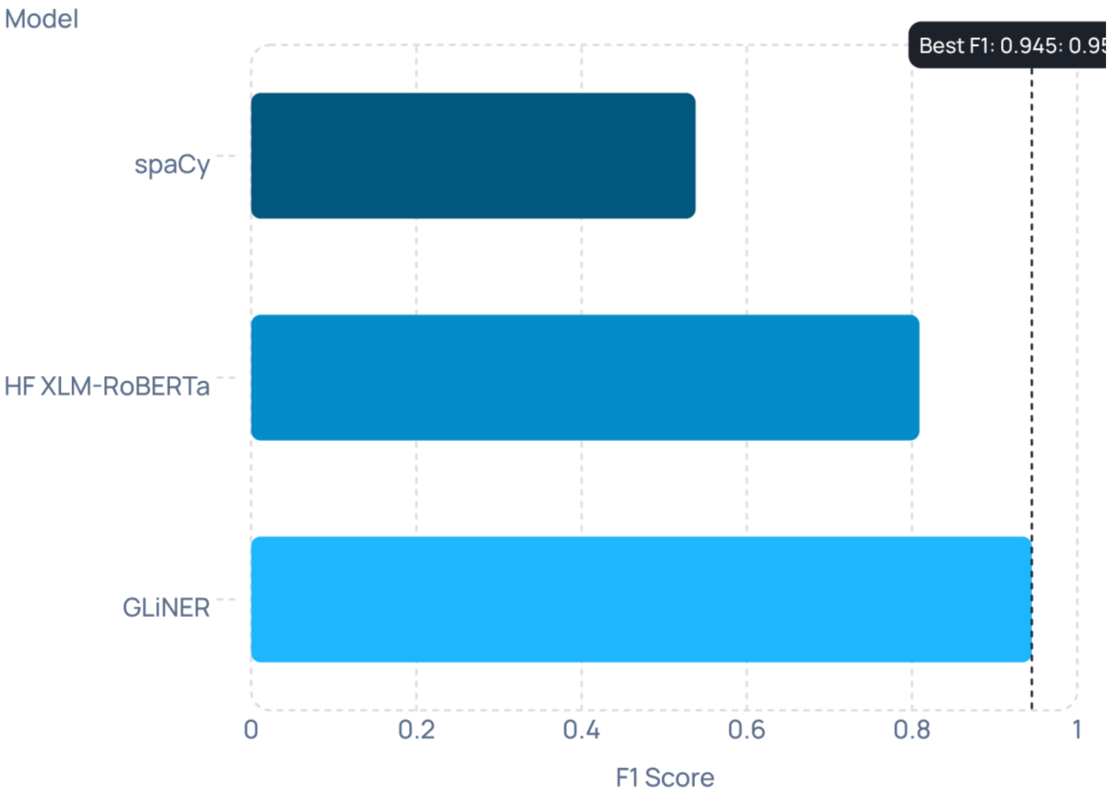
1. The predicted character span exactly matches the ground-truth span
2. The normalized entity type is correct

TP — True Positive
Correctly detected PII (span + type)

FP — False Positive
Non-PII incorrectly flagged

FN — False Negative
Real PII missed by detector

Benchmark Results



F1 Score Comparison

GLiNER achieves the strongest detection performance at **F1 = 0.945**. HF XLM-RoBERTa is competitive at 0.809. spaCy trails at 0.538 but offers a significant speed advantage.

End-to-end (spaCy): 67/97 entities correctly anonymized (69.1% rate), 2.8% exact output match, 0 remaining PII.

GLiNER

F1: 0.945 · 0.057 s/ex

HF XLM-R

F1: 0.809 · 0.024 s/ex

spaCy

F1: 0.538 · 0.007 s/ex

Conclusion & Future Work

What Was Achieved

A complete, modular Python pipeline for PII detection and anonymization across structured and free-text clinical data. GLiNER achieved the strongest detection performance in our benchmark (F1 0.945), while spaCy offers a lightweight alternative for latency-sensitive deployments.

Current Limitations

- Regex-based structured detection is brittle across format variations
- NER span errors propagate to anonymization output
- Free-text coverage limited to predefined entity types

Future Directions

→ Improve Detection Accuracy

Fine-tune NER models on domain-specific clinical text; expand regex patterns.

→ Expand Free-Text Support

Add multilingual support and custom entity types for broader clinical contexts.

→ Reasoning Extensions

Explore LLM-based retrieval, knowledge synthesis, and reasoning over anonymized datasets.